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Studierichting: Informatica
Oefeningenreeks 5G nr. 2.2

$$f(x) = -x^2 + 5x - 4$$

$$D = 9 \Rightarrow x_1 = \frac{-5+3}{-2} = 1 \text{ en } x_2 = \frac{-5-3}{-2} = 4$$

$$V = \pi \int_1^4 (-x^2 + 5x - 4)^2 dx$$

$$= \pi \int_1^4 (x^4 - 10x^3 + 33x^2 - 40x + 16) dx$$

$$\Rightarrow V = \pi [x^4 - 10x^3 + 33x^2 - 40x + 16]_1^4$$

$$= \frac{128\pi}{10} - \frac{47\pi}{10} = \frac{81\pi}{10}$$

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References